

Indian Institute of Technology, Bhilai



CSL351: Computer Networks Assignment - 11 (DHCP, DNS & WebServer)

Full Marks: 30

Deadline: 26/04/2026, Time: 11:59 PM

Submission Instructions:

- 1) Answer all the questions.
- 2) Deliverables in a .zip:
 - a) Readable Report [**3 Marks for report quality**] enumerating steps followed with screenshots for each of the important steps.
 - i) Put the screenshots in the report for better clarity
 - ii) Explain everything that has been asked in your report and highlight the important parts, and follow the instructions given in the question carefully.
 - b) The .pkt file of the CISCO packet tracer.
- 3) You can use LaTeX/Word to make the PDF.
- 4) The naming of the zip file should strictly follow the given format:
<Roll_No>_<Name>_<Assignment No>. If your name is Alex, your roll number is B23CS055, then the filename should be: B23CS055_Alex_11.zip
- 5) Any plagiarism case will be considered an unethical practice, and appropriate action will be taken against them.

Objectives:

1. To implement automatic IP address allocation using DHCP in a network.
 2. To configure DNS for resolving domain names to IP addresses.
 3. To set up and host a basic web server using HTTP.
 4. To enable seamless communication between clients and network services.
 5. To verify network functionality using basic commands and browser access.
-

A new digital campus system is being introduced by your college so that students can:

- Link their computers to the campus LAN or WiFi.
- Obtain network access automatically
- Students should be able to access the college website using a domain name.

The IT division is dealing with serious problems:

- IP conflicts occur when students manually configure their IP addresses.
- Many students are unable to access the website because they do not know the server's IP address.

The college has made the decision to implement a centralized network infrastructure utilizing:

- IP allocation that is automatic
- Access based on domain names
- Internal web hosting

You are assigned as a student network engineer to design this system in Cisco Packet Tracer.

System Requirements:

- Student PCs (multiple)
- 1 Switch
- 3 Servers:
 - DHCP Server
 - DNS Server
 - Web Server

Task to do:

- Fix IP Allocation Problem
- Enable Easy Website Access
- Launch A Demo Website

Part A : Topology Design

[7 Marks]

1. Place Devices
 - 3 Servers
 - 1 Switch
 - 6 PCs
2. Connect PCs to the Switch via **Copper Straight Through**
3. Connect the Switch to all servers using **Copper Straight Through**

Part B : Configuration**[10 Marks]**

1. Configure all PCs via DHCP.
2. Configure all Servers with Static IP.
3. Assign IP and name mapping in the DNS Server correctly
4. Configure the Webserver Demo Website to work with both the IP address and the domain name.

Part C : Verification**[10 Marks]**

1. Show that all PCs receive IP configuration dynamically using DHCP (IP, subnet mask, default gateway)
2. Ping between at least two PCs using:
 - a. IP address
 - b. Domain name
3. Access the web server:
 - a. Using IP address
 - b. Using domain name
4. Demonstrate that DNS resolution is working correctly.

Note: For each configuration and verification, you must explain the process and attach the corresponding screenshots to the report.